Background

XYZ Insurance

XYZ Insurance, is a leading provider of insurance services in Ireland, dedicated to providing our clients with peace of mind and financial protection. With a commitment to integrity and customer satisfaction, we strive to deliver innovative solutions tailored to meet the diverse needs of our policyholders. Our experienced team of professionals works tirelessly to ensure that claims are processed efficiently and fairly, upholding the highest standards of trust and reliability in the industry.

XYZ Insurance is facing a significant challenge with an increase in fraudulent claims, which is impacting both our financial performance and operational efficiency. The current process of investigating fraudulent claims is heavily reliant on manual efforts, consuming a substantial amount of time and resources from our insights team. This manual approach not only slows down the claims processing but also diverts valuable resources away from other critical tasks, hampering overall team productivity.

Furthermore, the sheer volume of fraudulent claims overwhelms our capacity to identify and address them effectively. Without a more streamlined and data-driven approach, we risk not only financial losses but also reputational damage due to our inability to detect and prevent fraudulent activities promptly.

In response to these challenges, XYZ Insurance recognizes the urgent need to the efficiency of their Quality and Risk teams. XYZ's executive team have contacted your team to urgently complete a Proof of Concept to identify common features which may be indicators of potential fraud, which will better enable their team to pinpoint claims for manual review.

The executive team are not strangers to analytical models, and while they would like the model to be easily explained to the Quality and Risk teams, their primary goal is to use this Proof of Concept to showcase potential uses and effectiveness of multivariate analysis and advanced analytics.

As well as delivering this Proof of Concept, the executive team are very keen to understand any other recommendations from your team on how they can leverage the wealth of information and data at their disposal to enhance decision-making processes and improve. XYZ Insurance want to reinforce their position a leader in the marketplace and understand that, in the current climate, they must adopt data and AI and make it a central part of their business.

Current XYZ Insurance Investigation Process

Currently, XYZ Insurance employs a series of manual checks and processes to investigate fraudulent claims, which include:

1. Documentation Review: Claims investigators meticulously examine the documentation submitted with each claim, scrutinizing details such as medical records, police reports, witness statements, and any other relevant paperwork. Discrepancies or inconsistencies in the documentation may raise red flags and trigger further investigation.
2. Interviews and Interrogations: Investigators conduct interviews with claimants, witnesses, and any other parties involved in the incident to gather additional information and assess the credibility of the claim. Through careful questioning and observation, they aim to uncover any signs of deception or dishonesty.

3. Data Analysis: While primarily a manual process currently, some basic data analysis is conducted to identify patterns or anomalies in claims data that may indicate potential fraud. However, this process is limited by the resources and expertise available, often leading to missed opportunities for detecting fraudulent activity.

Overall, these manual checks are time-consuming and resource-intensive, relying heavily on the judgment and expertise of claims investigators. While effective to some extent, they are prone to human error and limited in their ability to handle the increasing volume and complexity of fraudulent claims. Therefore, there is a pressing need for XYZ Insurance to adopt more advanced analytics and technology-driven solutions to enhance their fraud detection capabilities and improve operational efficiency.

Project Requirements

This project has two equally important parts:

1. Data Analysis and Analytical Modelling
2. Business Value & Recommendations

In Week 6, you will be asked to present your projects to a judging panel and an audience of your peers. The judging panel will not have seen your project work week-on-week and may not be familiar with the exact project that you are trying to solve. It is therefore recommended that you cover the following in your presentation:

- Client situation and objective
- Analytical approach taken and why (model used, rationale and logic for not using something else)
- High level model outputs and recommendations for how the client can use/implement.

Your team will have 15 minutes including Q&A to present your solution. You are free to choose the medium that you use to present- Accenture will provide a screen, anything else you need to bring yourselves. However, we recommend using a digital format for any visualisations (PowerPoint slides, R-slides, Tableau or PowerBI dashboards can be some examples), so that they are easily visible to your audience.

Remember! As a Data Scientist you are the translator of the data into business language and useful business insight.

Data

XYZ Insurance stores data about each claim that they have received including data on the policy, policy holder and vehicle in a single big table. The table also contains an indicator if the claim was found to be fraudulent or not, recorded by the Risk team.

For the purposes of this POC, XYZ Insurance have provided:

- Production data for last year with approximately 20,000 records,
- Data dictionary
- Information on the cost of manual review

Support

As part of the project clinic each week, you will be assigned a project mentor and have access to the client subject matter expert (SME) and data science SME.

The project mentor will act as a project lead and help you determine your tasks and project plan for each week. You will have 10 minutes of time each week with the Client SME and Data Science SME (20 mins total each week)

The Client SME will be able to answer questions about the client background and claim investigation process and help to clarify what the client is looking for from the solution you present. The Data Science SME will be able to answer questions about the methodologies and suggest techniques. It is suggested that you prepare your questions before consulting with the SMEs as they will be splitting their time across teams.

In Week 5, Storytelling experts will be available during the project clinic, to answer presentation questions.

Sample Project Flow

There are may different ways to deliver what the client requires. You may use Excel, Python, R or any analytics tools that you are comfortable with the perform your analysis.

The project flow below is intended to provide some helpful points – it is not a complete or mandatory checklist.

1. Clean the data: Assess the quality of the data, try to remove the redundant features and decide how to treat missing values if they exist.
2. Analyse the data using a common statistical tool: Visualise the data and perform univariate, bivariate and multivariate analyses as applicable.
3. Perform root cause analysis using advanced analytics techniques.
4. Decide on the two or three key findings from your solution. You should be able to explain why you took the approach you did, the benefits/savings to be gained by employing this model.
5. Talk about the next steps that can be taken by the company to improve their business such as:
    a. Are there any future insights not strictly relevant to the fraudulent claims which may be of interest to XYZ Insurance? Suggest how these insights can be applied to other areas of the business.
    b. Consider what other data XYZ Insurance, as an insurance provider, collects. Surplus to the data already provided to Accenture by XYZ Insurance, what other internal and external data sources could be utilised for any future analytical purposes (risk or otherwise)

You will present results from the analysis as well as your recommendation on other potential analytics use cases to XYZ Insurance. Remember that two or three key findings is better than a lot of irrelevant information. You may include graphs and/or statistics. Tell the story of the data and explore how variables may be relevant to each other, how the models and the data XYZ Insurance already collects can be utilized in solving the current problem ad applied to future ones.